

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Period: \_\_\_\_\_

1. Difference table.

| $x$ | $y$ | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ |
|-----|-----|------------|--------------|--------------|
| 0   | 2   | —          | —            | —            |
| 1   | 6   | 4          | —            | —            |
| 2   | 16  | 10         | 6            | —            |
| 3   | 38  | 22         | 12           | 6            |
| 4   | 78  | 40         | 18           | 6            |
| 5   | 142 | 64         | 24           | 6            |

The **third** differences are constant, so this data is best modeled by a degree-**3** polynomial.

2. Context-to-model matching.

A  $\rightarrow$  **3**      B  $\rightarrow$  **1**      C  $\rightarrow$  **2**      D  $\rightarrow$  **1**

Teacher Note

B and D are both linear, but they match to model type 1 independently. Profit per unit being constant (D) is also linear. C is quadratic because projectile height as a function of horizontal distance follows  $h = ax^2 + bx + c$ .

3. Assumption articulation.

- (a) **This model assumes the cyclist travels at a constant (unchanging) speed throughout the trip.**  
 (b) Domain:  $t \geq 0$  Reason: **Time cannot be negative; the cyclist starts at  $t = 0$ .** .....

4. Piecewise model description.

Piece 1 (when  $0 < h \leq 1$ ):  $C = 3.00$  Type: **constant**

Piece 2 (when  $1 < h \leq k$ , where  $3 + 1.5(h - 1) < 15$ ):  $C = 3.00 + 1.50(h - 1)$  Type: **linear**

Piece 3 (when  $h > k$  for the maximum threshold):  $C = 15.00$  Type: **constant**

Teacher Note

The daily cap is reached when  $3 + 1.5(h - 1) = 15 \Rightarrow h = 9$  hours. So Piece 2 applies for  $1 < h \leq 9$  and Piece 3 for  $h > 9$ . Accept reasonable interval notation.

5. Model selection with justification.

| $r$ | $A$ | $\Delta A$ | $\Delta^2 A$ |
|-----|-----|------------|--------------|
| 1   | 3   | —          | —            |
| 2   | 12  | 9          | —            |
| 3   | 27  | 15         | 6            |
| 4   | 48  | 21         | 6            |

Model type: **Quadratic**

Context clue: **Area is a 2-D (flat surface) measurement, which matches the  $\text{area} \propto r^2$  shape of a circle or spreading region.** .....

**Extension**   optional

**Extension (optional).**

| $x$ | $y$ | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ |
|-----|-----|------------|--------------|--------------|
| 0   | 0   | —          | —            | —            |
| 1   | 2   | <b>2</b>   | —            | —            |
| 2   | 8   | <b>6</b>   | <b>4</b>     | —            |
| 3   | 18  | <b>10</b>  | <b>4</b>     | <b>0</b>     |

The student’s claim is partially correct but imprecise. With 4 points, a degree-3 polynomial is the maximum needed; however, the second differences are already constant (all 4), so the data is quadratic (degree 2). A cubic is not required. The unique polynomial of *lowest* degree through these 4 points is actually quadratic ( $y = 2x^2$ ). .....